

I am currently a Senior Engineer of Kangaroo Network Group at Alibaba Cloud. My job focuses on improving networking foundations for cloud native systems and agentic sandboxes, aiming to provide flexible and efficient networks within containers and above cloud VPCs. I received my Ph.D. degree in computer science in 2022 from Northwestern University, under the guidance of Prof. Yan Chen.

Education

- **Ph.D. in Computer Science** 2017 – 2022
Northwestern University Evanston, IL
- Advisor: Prof. [Yan Chen](#) Area: Cloud Networks, Network Protocols GPA: 3.97/4.00
- **Master's in Computer Science** 2014 – 2017
Xi'an Jiaotong University Shaanxi, China
- Advisor: Prof. [Chengchen Hu](#) Area: SDN Rank: 1st/89 GPA: 3.81/4.00 Average: 91.7
- **B.E. in Software Engineering** 2010 – 2014
Xi'an Jiaotong University Shaanxi, China
- Rank: 1st/78 GPA: 3.94/4.00 Average: 92.4 (in Junior and Senior years)

Work Experiences

- **Alibaba Cloud** Shanghai, China
Senior Engineer, at Kangaroo Network Group Sep. 2022 – Present
- Aiming to provide flexible, efficient networks: within containers, above VPCs, for agentic sandboxes
- **SRI International** Menlo Park, CA
Senior Research Intern, at Computer Science Laboratory Summers of 2020, 2021
- Advisor: Dr. [Vinod Yegneswaran](#) and Dr. [Phillip Porras](#)

Research: Cloud Networks & SDN

- **Network Optimizing for Serverless Cloud** 2020 – 2024
- Serverless architectures bring flexibility but introduce network communication delay
- Seamlessly integrated **QUIC** into **OpenFaaS/K8s**, reduced cloud network latency by up to 40%
- Designed a clean-slate serverless network architecture for direct function chain communications
- **Routing Policy for Solving Reactive Model Overhead of SDN** 2016 – 2017
- Proposed a routing policy to reduce the SDN control channel bandwidth consumption by up to 80%
- Implemented in the **Floodlight** controller under **OpenFlow/P4** switch and **Open vSwitch**

Research: Formal Methods for Network Protocols

- **Formal Verification and Vulnerability Detection of LTE/5G Protocols** 2019 - 2020
- Used **TLA+** to formally specify the emergency call systems in **4G/5G** cellular network protocols
- Discovered serious availability and security issues in real world, **acknowledged by major carriers**
- **Formal Secure Configuration Search for Network Protocols** 2020 - 2022
- We converted formal verification into a search problem. Given the model and the properties, we search the boundaries of the configuration space where the system is always secure and reliable

Selected Publications

- You Li*, **Kaiyu Hou***, Hai Zhou, Yan Chen (*equal contribution), *Property Guided Secure Configuration Space Search*, International Conference on Information Security, 2024 (ISC'24)
- **Kaiyu Hou**, Sen Lin, Yan Chen, Vinod Yegneswaran, *QFaaS: Accelerating and Securing Serverless Cloud Networks with QUIC*, ACM Symposium on Cloud Computing, 2022 (SoCC'22)
- You Li*, **Kaiyu Hou***, Yan Chen, Hai Zhou (*equal contribution), *Property Guided Secure Configuration Space Search*, Under Review, Formal Methods in Computer-Aided Design, 2022 (FMCAD'22)
- **Kaiyu Hou***, You Li*, Yinbo Yu, Yan Chen, Hai Zhou (*equal contribution), *Discovering Emergency Call Pitfalls for Cellular Networks with Formal Methods*, ACM International Conference on Mobile Systems, Applications, and Services, 2021 (MobiSys'21)
- Xiaochun Wu, **Kaiyu Hou**, Xue Leng, Xing Li, Yinbo Yu, Bo Wu, Yan Chen, *State of the Art and Research Challenges in the Security Technologies of Network Function Virtualization*, Internet Computing, 2020
- Yinbo Yu, You Li, **Kaiyu Hou**, Yan Chen, Hai Zhou, Jianfeng Yang, *CellScope: Automatically Specifying and Verifying Cellular Network Protocols*, ACM SIGCOMM, 2019, Poster (SIGCOMM'19)
- Xue Leng, **Kaiyu Hou**, Yan Chen, Kai Bu, Libin Song, *SDNKeeper: Lightweight Resource Protection and Management System for SDN-based Cloud*, IEEE/ACM International Symposium on Quality of Service, 2018 (IWQoS'18)
- Chengchen Hu, **Kaiyu Hou** (1st student author), Hao Li, Ruilong Wang, Peng Zheng, Peng Zhang, Huan Zhao Wang, *SoftRing: Taming the Reactive Model for Software Defined Networks*, IEEE International Conference on Network Protocols, 2017 (ICNP'17)
- Xiuwen Sun, **Kaiyu Hou**, Hao Li, Chengchen Hu, *Towards A Fast Packet Inspection over Compressed HTTP Traffic*, IEEE/ACM International Symposium on Quality of Service, 2017 (IWQoS'17)

Awards & Honor

- **Terminal Year Cabell Fellowship**, Northwestern University 2021
- **Best Teaching Assistant Award**, Northwestern University 2020
- **Outstanding Graduate Award**, Xi'an Jiaotong University 2014, 2017
- **Excellent Student Award**, Xi'an Jiaotong University 2011, 2012, 2013, 2015, 2016
- **Google Excellence Scholarship**, Awarded to 3 students from each of 20 top Chinese universities 2013
- **Meritorious Winner**, Mathematical Contest in Modeling (MCM) 2013
- **Silver Medal**, ACM-ICPC China Province Contest 2012, 2013

Activities & Experiences

- **Reviewer/Sub-reviewer** of AISA-CCS (21'), CCS (18', 19'), ICDCS ('18), IEEE ToN
- **Student President of Computer Science Dept.** (Class 2017), Xi'an Jiaotong University 2014 - 2017
- **Chair, the ACM-ICPC Club**, Xi'an Jiaotong University 2012 - 2013

Skills

- Serverless Computing, Serverless Network Architectures, Cloud Networks, SDN, and P4
- QUIC Protocol, TCP/IP Protocols, 3GPP Cellular Network Protocols, and 802.11 Protocols
- Python, Go-lang | Container, Microservices, Kubernetes, and CNI
- Formal Methods for Network Protocols, TLA+, IC3